

SOFTWARE ENGINEERING PROJECT

FINGUARD AI

Real-Time Financial Fraud Detection System

PROTOTYPE DOCUMENT

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1 Prototype Overview

FinGuard AI is a real-time financial fraud detection system designed to evaluate transactions as they occur and produce an explainable risk decision. The prototype combines deterministic rules with supervised and unsupervised machine learning in a single scoring pipeline.

The basic prototype flow is:

**Transaction → Authentication → Validation → Feature Extraction → Fraud Detection
→ Risk Score → Decision**

The system is intended to reduce detection delay, improve fraud detection quality, reduce unnecessary false positives, and provide analysts with evidence explaining risky transactions.

2 Objectives

- Evaluate transactions in real time.
- Combine a rule engine, XGBoost, and Isolation Forest for hybrid fraud detection.
- Produce a final risk score between 0 and 100.
- Provide SHAP-based explanations for model-driven decisions.
- Provide an analyst interface for reviewing flagged transactions.
- Record analyst decisions for future model improvement.

3 Prototype Workflow

3.1 Transaction Processing

A customer submits a payment or transfer through the application. The API gateway authenticates the request using JWT and applies role-based access control. The transaction service validates and logs the transaction before extracting the features required for fraud detection.

The principal features include:

- Transaction amount
- Transaction velocity
- Geographic deviation from customer history
- Device fingerprint
- Other available behavioural signals

3.2 Fraud Detection

The extracted features are evaluated by three detection components:

Component	Prototype role
Rule Engine	Applies deterministic, hand-written fraud thresholds.
XGBoost	Produces a supervised probability-of-fraud signal.
Isolation Forest	Identifies unusual behavioural patterns and outliers.

The outputs are combined by a risk aggregation layer to generate a final score from 0 to 100.

3.3 Risk Decision

Risk tier	Score	Action
Low	0–30	Approve transaction.
Medium	31–60	Hold and request customer verification.
High	61–80	Hold transaction and alert analyst.
Critical	81–100	Block transaction and open investigation case.

4 Prototype Interfaces

4.1 Customer Interface

The customer interface provides a simple transaction form containing the required transaction details. After submission, the system displays the resulting transaction status.

- Transaction amount and recipient details
- Submit transaction control
- Transaction status
- Verification message when additional confirmation is required

4.2 Analyst Dashboard

The analyst dashboard displays flagged transactions and allows an analyst to open an individual case. A case contains:

- Transaction identifier and amount
- Final risk score and risk tier
- Rule, XGBoost, and anomaly signals
- Relevant transaction and behavioural features
- SHAP feature contributions
- Current case status
- Analyst decision

The analyst can record one of the following outcomes:

Confirm Fraud False Positive Escalate Case

5 Explainability and Feedback

SHAP is used to provide per-transaction feature attribution for model-driven decisions. Instead of displaying only a risk score, the prototype presents the features that contributed to the decision, such as transaction amount, device information, location deviation, velocity, and account history.

Analyst decisions are stored in a decision feedback database. Confirmed fraud and false-positive outcomes can subsequently support relabelling and future model retraining.

Detection → Analyst Review → Verdict → Feedback Database → Future Retraining

6 Technology Stack

Layer	Proposed technology
Authentication	JWT and role-based access control
Backend	REST API and transaction/scoring services
Supervised ML	XGBoost
Unsupervised ML	Isolation Forest with scikit-learn
Explainability	SHAP
Database	Relational or document database
Frontend	Web-based customer and analyst interfaces

7 Evaluation

The prototype will be evaluated using the project's defined KPIs:

- **Fraud Recall:** proportion of actual fraud correctly flagged.
- **False Positive Rate:** proportion of legitimate transactions incorrectly flagged.
- **Mean Time to Detect:** latency between transaction occurrence and risk-score generation.
- **Analyst Productivity:** change in average investigation time using the dashboard.
- **Fraud Loss Avoided:** estimated monetary value of prevented fraud.

8 Prototype Deliverables

The prototype will provide:

1. Transaction ingestion with JWT authentication and RBAC.
2. Rule engine and baseline XGBoost fraud scoring.
3. Risk aggregation with a 0–100 score.
4. Isolation Forest anomaly detection.
5. Basic analyst dashboard for flagged transactions.
6. SHAP-based explanations.

7. Analyst decision and feedback storage.
8. KPI reporting for system evaluation.

9 Conclusion

The FinGuard AI prototype provides a concise implementation of the proposed fraud-detection lifecycle. It combines deterministic rules, supervised and unsupervised machine learning, risk aggregation, explainability, and analyst feedback into a single system. The prototype therefore provides the foundation for implementing and evaluating the complete FinGuard AI system.